McKinsey & Company

Social Sector Practice

Overcoming sub-Saharan Africa's health workforce paradox

Many countries in sub-Saharan Africa face high healthcare worker unemployment—while coping with a severe shortage of healthcare professionals. Five strategic shifts may help narrow the gap.

by Tania Holt and Ying Sunny Sun
with Marilyn Kimeu and Parusha Raidoo

November 2024

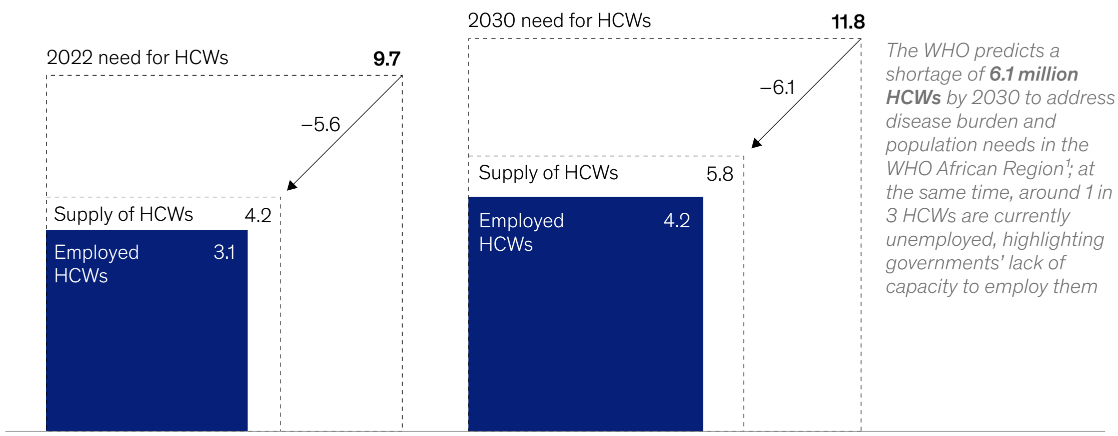
Mustering adequate human resources for healthcare is a significant challenge for most health systems globally. Everywhere you look, countries are facing shortages in qualified healthcare personnel, uneven distribution of workers, inadequate training, and challenging work environments.[1] These issues are most acute in low- and middle-income countries, particularly in Africa, which carries an estimated 25 percent of the world's disease burden but accounts for only 3 percent of the global healthcare workforce.[2] Many trained healthcare workers (HCWs) remain unemployed because health ministries face budget constraints and other competing priorities, giving a false indication of oversupply.[3]

The World Health Organization (WHO) projects a 6.1 million HCW shortage in its African Region,[4] based on the needs of the population and disease prevalence (equating to approximately 60 percent of universal health coverage).[5] This shortfall is compounded by a paradox: the region also faces high unemployment—approximately one in three HCWs are unemployed (Exhibit 1).[6] This means that about 1.5 million trained HCWs could be missing

Exhibit 1

Sub-Saharan Africa faces a severe shortage of healthcare workers, despite widespread unemployment in the sector.

WHO African Region[1] baseline and forecasted health workforce estimates,[2]
millions of healthcare workers (HCWs)



Note: Figures may not sum, because of rounding.
[1]The WHO African Region includes all countries in sub-Saharan Africa as used in the statistics of United Nations institutions, except for Sudan and Somalia (excluded) and Algeria (included).
[2]Physicians, nurses, midwives, dentistry personnel, pharmaceutical personnel, laboratory health workers, environmental and public-health workers, community and traditional health workers, health management and support health workers, and other health workers (which include medical assistants, dieticians, nutritionists, occupational therapists, medical imaging and therapeutic-equipment technicians, optometrists, ophthalmic opticians, physiotherapists, personal-care workers, speech pathologists, and medical trainees).
Source: "Needs-based health workforce requirements to address Africa's disease burden and demographic evolution," WHO African Region, 2024

McKinsey & Company

1 "Health workforce," World Health Organization (WHO), accessed September 2024.
2 "COVID-19 Management: Curriculum for Community Health Workers," Africa CDC, November 2020.
3 "Road map for scaling up the human resources for health: For improved health service delivery in the African region 2012-2025," WHO, 2013.
4 The WHO African Region includes all countries in sub-Saharan Africa as used in the statistics of United Nations institutions, except for Sudan and Somalia (excluded) and Algeria (included).
5 *Needs-based health workforce requirements to address Africa's disease burden and demographic evolution: Implications for investing in the education and employment of health workers, 2022–2030*, WHO African Region, 2024.
6 *A decade review of the health workforce in the WHO African Region, 2013-2022*, WHO African Region, 2024.

from the region's health workforce in 2030 despite the substantial unmet healthcare needs.

Considering escalated disease prevalence, reduced productivity, and lack of resources, the HCW shortage could potentially be 93 percent larger.[7] Regardless of the extent of this gap, addressing it could support the region in reducing the disease burden (measured in disability-adjusted life years, or DALYs) by 35 percent and increasing the region's GDP by an estimated $22 billion a year (Exhibit 2).

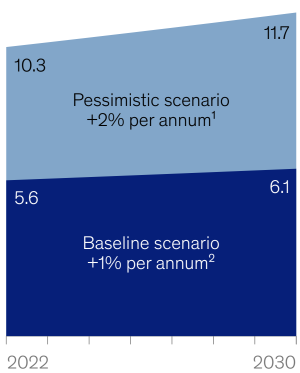
Adding to the problem, the region is facing a "brain drain" of healthcare professionals; many are seeking better opportunities and working conditions abroad, depleting the pool of skilled personnel needed to maintain and improve health systems. One in every ten doctors and nurses are already working abroad, while on average, 42 percent of HCWs have intentions to migrate to another country in the future, looking mostly to Australia, Canada, the United Kingdom, and the United States.[8]

Exhibit 2

Filling the health workforce gap could reduce disease burden and increase GDP in the region.

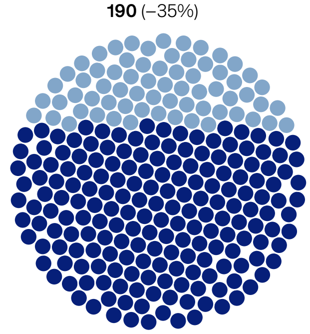
Healthcare worker (HCW) requirements in Africa

Estimated healthcare worker shortage in the WHO African Region, millions

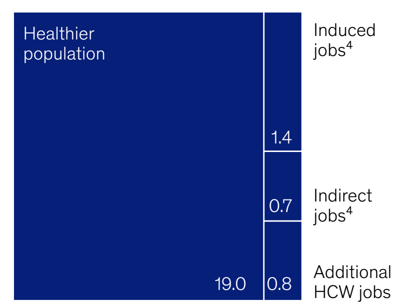


Gains from filling the 6.1 million employee shortfall in 2030[3]

Disability-adjusted life years saved, millions of years



GDP increase, $ billion



Note: Approaches used to calculate figures: disability-adjusted life years (DALYs) modeled by calculating reduced DALYs due to increased universal health coverage (UHC), based on exponential relationship between DALYs and UHC; healthier population impact modeled by calculating reduced DALYs from the working population due to increased UHC, then multiplying this by labor statistics to estimate economic benefits; impact from additional HCW jobs is economic productivity of the health workforce (based on the proportion of productivity attributed to the health sector), multiplied by the additional HCWs needed by 2030; indirect and induced jobs impact estimated by applying a job multiplier to direct HCW jobs, multiplying this by GDP per employed person and a value-add multiplier; indirect jobs produce goods and services for direct job workers, while induced jobs arise from the personal spending of direct and indirect workers. Figures may not sum, because of rounding.
[1]Scenario showing escalated disease prevalence and a maximum time requirement from health workers (assuming low competencies and lack of logistics to work).
[2]Scenario showing baseline disease prevalence and an average time requirement from health workers (assuming normal competencies and logistics to work).
[3]To cut the current 6.1 million shortage in half, an investment of $120.4 billion (less than 2% of the region's GDP) is likely required over 8 years, with roughly 19% going toward training health workers and 81% toward keeping existing jobs and hiring new graduates.
[4]Indirect jobs exist to produce the goods and services needed by the workers with direct jobs; induced jobs are created by the additional personal spending by both direct and indirect workers.
Source: *Global burden of disease 2021: Findings from the GBD 2021 study*, Institute for Health Metrics and Evaluation, 2024; "Global Health Observatory indicator metadata registry list: Disability-adjusted life years (DALYs)," WHO; "Needs-based health workforce requirements to address Africa's disease burden and demographic evolution," WHO African Region, 2024

McKinsey & Company

7 Ibid.
8 Ibid.

The healthcare workforce also faces challenges with equity, particularly in the distribution and quality of jobs, as well as gender disparity. Many healthcare roles fail to meet the International Labour Organization's standards for "decent work," leading to high attrition rates due to poor working conditions.[9] Gender disparities heighten equity concerns in the healthcare workforce. For instance, in sub-Saharan Africa, about 70 percent of community HCWs are young women, most of whom are unpaid, with only 43 percent receiving nonmonetary incentives and just 23 percent receiving stipends. Globally, while women make up 70 percent of the health workforce, they occupy only about 25 percent of senior roles. Similarly, in Africa, just 38 percent of ministries of health are led by women, highlighting the persistent gender inequity in leadership positions within the sector.[10]

A reimagined healthcare workforce ecosystem—and a call to action

Sub-Saharan Africa faces an urgent need to reimagine its strategies to safeguard and enhance the healthcare workforce. In this article, we set out five shifts that move away from outdated care delivery and workforce models. These shifts aim to create a high-quality, cost-effective, and adaptable HCW model that can be tailored to the needs and dynamics of each population. Embracing these shifts can help national health services address the HCW shortage and foster a more productive and healthier society.

However, proactive commitment at the highest levels of leadership is crucial to ensure meaningful impact. Leaders may need to champion creating equitable, inclusive, and high-quality healthcare jobs, supported by institutional frameworks, policies, and investments that foster favorable working conditions. Such investments in the development of the healthcare workforce are essential for driving all five shifts and establishing high-quality, cost-effective healthcare systems that are sustainable in the long term.

Shift 1: Plan the workforce strategically

Many sub-Saharan countries plan their workforce using overly broad standards, typically based on population or facilities, and don't have an accurate count of HCWs or their geographical distribution, skill mix, and productivity. Further, they often have narrow and rigid definitions of roles that often do not include experts such as nurse practitioners, health officers, and community health workers. The following two strategies could deliver better workforce configurations and free up worker capacity.

Deploy data analytics and digital tools. Technology-enabled strategic workforce planning can help policy and decision makers identify new workforce configurations. Better deployment of workers can help countries take advantage of advances in diagnostics, therapeutics, and personalized care; drive productivity; prepare for surges during outbreaks; and maximize potential synergies across core and health-enabling roles. For example, a planning and allocation tool from Touch Health allocates HCWs to health facilities where they will have the biggest impact on patient care, within budgetary constraints. The system has been adopted in five countries across 10,000 hospitals and health facilities, optimizing 39,000 health workers.

Structure roles and allocate tasks within the local context. Several countries have adopted task shifting and established new, localized workforce models that allocate tasks to nontraditional HCWs. At one such center in Uganda, the shift to community-based birthing centers reduced perinatal mortality by 87 percent, nearly doubled the number of deliveries attended by skilled HCWs, and boosted antenatal care coverage by about 90 percent.[11] However, these initiatives are often limited in scope, lack sustainability, and are typically confined to addressing specific, isolated cases.

The scope for such programs can be broadened to new populations and to address a growing

[9] Delphin Kolié, Remco Van De Pas, Laurence Codjia, and Pascal Zurn, "Increasing the availability of health workers in rural sub-Saharan Africa," *Human Resources for Health*, 2023, Volume 21, Number 20.
[10] "Delivered by women, led by men: A gender and equity analysis of the global health and social workforce (Human Resources for Health Observer Series, No. 24)," WHO, March 14, 2019.
[11] "Ot Nywal Me Kuc Birth Centre," Mother Health International, accessed September 2024.

burden from noncommunicable diseases. A reconfiguration of roles and tasks could aim to build multidisciplinary teams of professionals delivering integrated care at the top of their licenses.[12] This could include strengthening and professionalizing community health workers (CHWs) and home-based roles, further specializing highly skilled roles (such as doctors), enhancing competencies in certain roles to give workers greater autonomy (such as nurse practitioners and prescribers), and establishing new professions for targeted services (for example, community-based midwives, public-health physicians, and clinical-administrative-support workers).

Shift 2: Reform healthcare worker training

In 2022, approximately 4,000 institutions across sub-Saharan Africa produced only about 255,000 healthcare graduates.[13] Boosting this capacity and integrating innovative training methods is essential to swiftly closing the talent gap. Additionally, to ensure the quality of care is upheld, WHO is developing competency-based curricula that countries can use to benchmark their local training programs.[14]

The COVID-19 pandemic prompted sub-Saharan African countries to explore flexible, digitally enabled, and cost-effective training methods for frontline HCWs. For example, in the Democratic Republic of the Congo, training delivered by mobile phones helped 5,000 HCWs deliver vaccines efficiently and safely.[15] A STEM-focused learning platform in Nigeria leverages AI to create personalized, engaging lessons for out-of-school rural students, integrating national context and nuances to enhance local comprehension.

Not every training need can be addressed with a mobile app, but it is possible to build on these early successes and design fit-for-purpose training that emphasizes flexible formats, competency-based mastery of in-demand skills, and individualized learning journeys. For example, in 2020, ICAP at Columbia University partnered with Resolve to Save Lives and 11 ministries of health to design an in-service training program for frontline workers, tailored to the needs of different regions. In about four months, the competency-based program trained an estimated 9,000 HCWs across 1,000 facilities using a combination of in-person, virtual, and self-guided sessions.[16]

Shift 3: Increase entrepreneurial ventures

Sub-Saharan Africa faces a stark imbalance between available jobs and supply of HCWs. An estimated 85 percent of HCWs are employed in the public sector, while almost one in three HCWs in the region are unemployed.[17] At the same time, there has been limited entrepreneurship and job creation within the private health sector, particularly in underserved communities. The private sector invests only minimally in sub-Saharan Africa's nascent health-enabling industries, which include local pharmaceutical manufacturing, digital healthcare, and health insurance. The following actions could help.

Partner with the private sector. Many countries have attempted to stimulate healthcare entrepreneurship through regulatory and policy changes. Several countries, including Ethiopia, Namibia, Nigeria, and Uganda, had already changed policies by 2007 to grant autonomy to nurses and midwives to establish their own private clinics.[18] However, HCWs often lack entrepreneurial and business management skills, as well as key resources (such as finance, talent, and mentors) and access to markets required for success.

In response, several promising models have emerged over the past decade to support HCWs'

---

12 "Top of license" practice means that healthcare professionals are working at the full extent of their training and skills, focusing on tasks that only they are qualified to perform, thereby maximizing efficiency and the quality of care provided.
13 *A decade review of the health workforce in the WHO African Region, 2013-2022*, WHO African Region, 2024.
14 McKinsey interview with WHO AFO health workforce team lead.
15 Gregory Kuzmak, "How new training approaches are transforming health workers' ability to deliver vaccines," Rockefeller Foundation, February 22, 2022.
16 Fatima Tsiouris et al., "Rapid scale-up of COVID-19 training for frontline health workers in 11 African countries," *Human Resources for Health, 2022*, Volume 20, Number 43.
17 Adam Ahmat et al.,"The health workforce status in the WHO African Region: findings of a cross-sectional study," *BMJ Glob Health*, May 2022, Volume 7; *A decade review of the health workforce in the WHO African Region, 2013-2022*, WHO African Region, 2024.
18 "Why policy matters: Regulatory barriers to better primary care in Africa," USAID, August 2007.

pursuit of entrepreneurial projects. For example, South Africa's Unjani Clinic, a not-for-profit organization, operates a network of about 200 primary healthcare "container clinics" in low-income communities, making it the largest nurse-led initiative in the country. Nurses obtain funding and training to purchase, equip, and operate container clinics over a five-year period, scaling to employ about five workers and achieving financial sustainability after approximately a year.[19] To date, the network has employed roughly 700 people and provided an estimated 5.2 million consultations to patients.

Develop health-enabling industries. Digital healthcare, pharmaceutical manufacturing, health insurance, and other enabling industries play crucial roles in ensuring resiliency, accessibility, and responsiveness. However, these industries are nascent in most countries in sub-Saharan Africa. For example, manufacturers on the continent produce only 1 percent of administered vaccines,[20] and more than 90 percent of medical devices in public hospitals across the continent are sourced from abroad.[21] In contrast, India's innovation ecosystem has created an enabling environment for the growth of its pharmaceutical-manufacturing industry, which is now the world's third-largest manufacturer by volume. The pharmaceutical sector in India employs an estimated 2.7 million people and has about 10,500 manufacturing units,[22] while Africa has approximately 375 pharmaceutical manufacturers.[23]

Shift 4: Manage talent proactively

Sub-Saharan Africa struggles to attract and retain healthcare workers, particularly in rural areas where 60 percent of the population resides.[24] For example, in rural areas of South Africa, Tanzania, and Kenya, which are home to approximately 46 percent, 70 percent, and 80 percent of each country's population, respectively, only 12 percent, 25 percent, and 16 percent of the countries' doctors provide care.[25] This uneven distribution and poor retention significantly weaken healthcare capacity in the region. Many healthcare workers leave the sector due to high levels of burnout and job dissatisfaction.[26]

Finding healthcare workers to serve rural communities is a long-standing challenge. Financing incentives may address these gaps: for example, in South Africa, rural education interventions have produced students with greater willingness and readiness for rural practice.[27] However, these initiatives often face sustainability challenges given their reliance on funding.[28]

Sub-Saharan Africa may need to rethink HCW incentives, scale community-based cadres and rural pipelines in underserved communities, and equip the workforce with appropriate resources, including digital and data-driven solutions that support health service delivery. For example, offering incentives[29] to provide care in rural areas has worked to some extent in Vietnam. Assistant doctors—midlevel

[19] "Expanding community healthcare," Mediclinic Southern Africa, September 28, 2023.
[20] "What will it take to develop a sustainable vaccine manufacturing ecosystem in Africa?," Africa CDC, October 2023.
[21] Geoffrey Banda et al., *The localisation of medical manufacturing in Africa*, Institute for Economic Justice, November 2022.
[22] "Sector: Pharmaceuticals," Invest India, accessed September 2024.
[23] "Getting quality medicines to patients faster in Africa: How to solve access issues," blog post by Mridu Bhutani, IQVIA, January 16, 2023.
[24] "Rural population data (% of total population) - Sub-Saharan Africa," World Bank Group, accessed September 2024.
[25] Nathanael Sirili et al., "Accommodate or reject: The role of local communities in the retention of health workers in rural Tanzania," *International Journal of Health Policy and Management*, January 2022, Volume 11; Grace Mburu and Gavin George, "Determining the efficacy of national strategies aimed at addressing the challenges facing health personnel working in rural areas in KwaZulu-Natal, South Africa," *African Journal of Primary Health Care & Family Medicine*, July 2017, Volume 9, Number 1; Kaja Momanyi et al., "Current status of family medicine in Kenya; family physicians' perception of their role," *African Journal of Primary Health Care & Family Medicine*, September 2020, Volume 12, Number 1.
[26] Benyam W. Dubale et al., "Systematic review of burnout among healthcare providers in sub-Saharan Africa," *BMC Public Health*, 2019, Volume 19, Number, 1247.
[27] S. C. Van Schalkwyk et al., "'Going rural': Driving change through a rural medical education innovation,'" *Remote Rural Health*, 2014, Volume 14, Number 2493; Elma de Vries and Steve Reid, "Do South African medical students of rural origin return to rural practice?," *South African Medical Journal*, October 2003, Volume 93, Number 10.
[28] "Performance based financing: Inspiring new approaches to public financial management in health and education," blog post by Srinivas Gurazada, Moritz Piatti, and Thomas Poulsen, *World Bank Blogs*, May 5, 2022.
[29] Incentives can include salary increases, promotions within in the year, and development opportunities.

primary care providers—are recruited from rural areas and undergo a two-year accredited program to provide basic primary care to the local population.[30] The assistant doctors were 70 percent more likely to move to rural areas than doctors when offered the same salary percentage increases, and twice as likely to move to rural areas when offered a promotion within a year.[31] Solutions to support workers' mental health and well-being may reduce attrition. For example, in South Africa, software company Dimagi has developed an AI coach for frontline HCWs that offers training, performance feedback, and encouragement.

Shift 5: Optimize productivity

Sub-Saharan Africa has struggled to realize the full potential of its health workforce due to suboptimal use of employees' time and skills, as well as inadequate performance management and supervision of healthcare workers. A study in Tanzania reports that HCWs show up for their assignments just 66 percent of the time, of which only 47 percent is considered productive time, while Kenya, Sierra Leone, and Malawi have absenteeism rates of 39 percent, 27 percent, and 18 percent, respectively.[32] Absenteeism, particularly in the afternoon when patient flow decreases, is a major contributor to nonproductive time.[33] Nonfinancial incentives[34] have sometimes enhanced staff motivation, but there remains considerable room to increase productivity and performance.[35]

Strategies to enhance healthcare delivery in sub-Saharan Africa could involve redesigning core processes to focus HCWs' time and skills on value-added activities. Digital solutions such as electronic health records, information exchange, and clinical-decision-support tools can be highly effective.

Leaders could also optimize clinical operations by aligning scheduling resources with patient flow and capacity and empower patients to manage their own care through digital therapeutics, self-care, and wellness initiatives. Digital tools can automate routine tasks, provide real-time clinical insights, and generate personalized care recommendations, potentially increasing HCW productivity in sub-Saharan Africa by up to 15 percent by 2030.[36] Examples of digital applications include the following:

— Rwanda has developed a digital health plan to improve public access to healthcare, including passing a law to foster partnerships with private providers of health services.

— Tanzania's GoTHOMIS, an interoperable health information system installed in about 10 percent of healthcare facilities, has already boosted productivity levels by centralizing facility records and automating clinical tasks such as scheduling and lab orders.[37]

— South Africa piloted a cloud-based AI solution to enable predictive analysis and operational intelligence to support access control, queue management, and appointment scheduling. The pilot ran for six months, managing 25,000 patients a month.[38]

— In Sierra Leone, a survey-based digital tool runs on smartphones and measures CHWs' performance, using data-driven insights and reducing the workload of supervisors.[39]

---

30 "Human resources for health country profiles: Vietnam," WHO, 2016.
31 Anh Thi Cam Pham, "Exploring determinants for recruitment and retention of family doctors for rural practice in Vietnam: Lessons from a discrete choice experiment," PhD diss., University of Texas at Dallas, December 2017.
32 *Harmonised health facility assessment (HHFA) 2018-2019* , Government of Malawi, Ministry of Health and Population, 2019.
33 *Assessment of health worker productivity in Tanzania*, Benjamin William Mkapa Foundation, September 2015.
34 Incentives can include certifications and training opportunities, career development, interfacility competitions, and living conditions.
35 Marjolein Dieleman and Jan Willem Harnmeijer, *Improving health worker performance: In search of promising practices*, WHO, 2006.
36 "How digital tools could boost efficiency in African health systems," McKinsey, March 10, 2023.
37 "Government of Tanzania Health Operations Management Information System (GoTHOMIS)," USAID, accessed September 2024.
38 "AI pilot at Limpopo Provincial Government Clinic," Mint South Africa and Limpopo Department of Health, accessed September 2024.
39 "Strengthening Sierra Leone's health system with a new digital supervision tool for community health workers," Last Mile Health, January 2024.

While addressing the healthcare workforce challenges in sub-Saharan Africa requires a comprehensive and multifaceted approach, it may not be realistic to implement all the shifts at once. To get started, leaders will need to commit to the pursuit of equitable, inclusive, and high-quality healthcare jobs. Each country will then need to prioritize changes based on its unique needs and most pressing challenges. From this starting point, countries can cascade and scale additional shifts. These investments in healthcare workforce development could help drive progress toward a more healthy and productive population as well as close sub-Saharan Africa's employment and productivity shortfalls.

Tania Holt is a senior partner in McKinsey's London office, **Ying Sunny Sun** is a partner in the Lisbon office, **Marilyn Kimeu** is an associate partner in the Nairobi office, and **Parusha Raidoo** is a capabilities and insights analyst in the Johannesburg office.

This article was edited by Rama Ramaswami, a senior editor in the New York office.

Designed by McKinsey Global Publishing